Exercise 15.7.47

Mass, Center of Mass, and Moment of Inertia: Integral Set-Up

Problem

Set up, but do not evaluate, integral expressions for (a) the mass, (b) the center of mass, and (c) the moment of inertia about the z-axis. The solid enclosed by the cylinder $y = x^2$ and the planes $z = 0$ and $y + z = 1$; $\rho(x, y, z) = \sqrt{x^2 + y^2}$.

The solid, the density, and the notation

The symbol E denotes the solid region in space. The symbol ρ denotes the density, that is, the mass per unit volume at a point of E. The symbol dV denotes the volume element of the solid; in rectangular coordinates $dV = dz\,dy\,dx$. The symbol m denotes the mass of E, the symbols $\bar{x}, \bar{y}, \bar{z}$ denote the coordinates of the center of mass, and I_z denotes the moment of inertia about the z-axis.

Three surfaces bound E:

- the cylinder $y = x^2$, a parabolic sheet that runs parallel to the z-axis;
- the plane $z = 0$, the floor;
- the plane $y + z = 1$, that is $z = 1 - y$, the ceiling.

The plane $y = 1$ is not a separate boundary. It is the line where the floor and the ceiling meet, because $z = 0$ and $y + z = 1$ give $y = 1$.

For a fixed point (x, y), a vertical line through (x, y) enters E at $z = 0$ and leaves E at $z = 1 - y$. Thus E is z-simple. The projection of E on the xy-plane is the plane region

$$D = \{(x, y) : -1 \le x \le 1,\ x^2 \le y \le 1\}.$$

The lower curve $y = x^2$ is the wall. The upper line $y = 1$ is the top of the wall, where the floor and the ceiling meet. The two curves intersect at $x = \pm 1$, so the range of x is $-1 \le x \le 1$. Figure 1 shows E, and Figure 2 shows D with one vertical column.

Why this order of integration is legal. The limits of x and y do not depend on z. The limits of y do not depend on z either. The limit of z depends on y only. Therefore we integrate with respect to z first, then with respect to y, then with respect to x:

$$\iiint_E f(x, y, z)\,dV = \int_{-1}^{1} \int_{x^2}^{1} \int_{0}^{1-y} f(x, y, z)\,dz\,dy\,dx.$$

Each inner integral has constant limits or limits that contain only the outer variables. This condition makes the description of E correct.

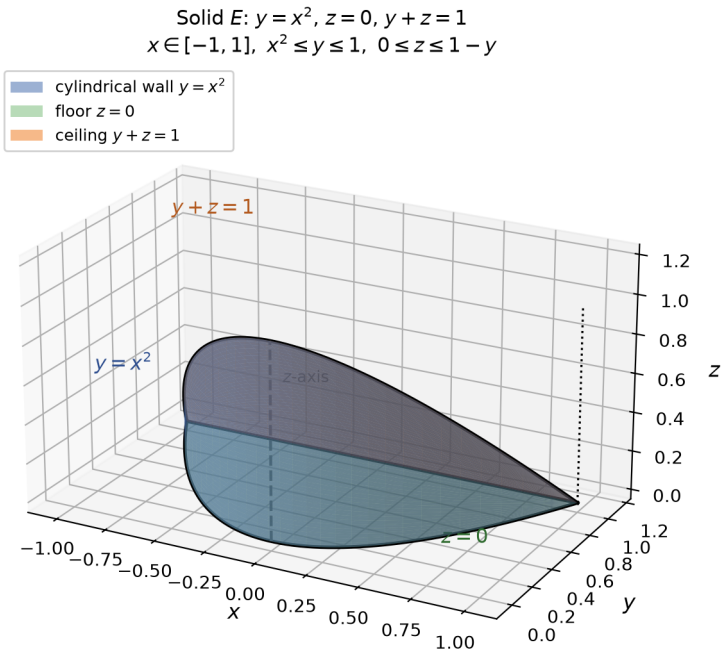


Figure 1: The solid E of Exercise 15.7.47. The blue sheet is the parabolic wall $y = x^2$. The green sheet is the floor $z = 0$. The orange sheet is the ceiling $y + z = 1$. The solid lies between the floor and the ceiling, and inside the wall.

Part (a): the mass

The mass is the integral of the density over the solid:

$$m = \iiint_E \rho(x, y, z)\, dV = \int_{-1}^{1} \int_{x^2}^{1} \int_{0}^{1-y} \sqrt{x^2 + y^2}\, dz\, dy\, dx.$$

The integrand does not contain z, so the inner integral is the length of the column times the density on that column:

$$\int_0^{1-y} \sqrt{x^2 + y^2}\, dz = \Big[z\sqrt{x^2 + y^2} \Big]_0^{1-y} = (1 - y)\sqrt{x^2 + y^2}.$$

Hence

$$\boxed{m = \int_{-1}^{1} \int_{x^2}^{1} (1 - y)\sqrt{x^2 + y^2}\, dy\, dx}$$

or, with all three integrations displayed,

$$\boxed{m = \int_{-1}^{1} \int_{x^2}^{1} \int_{0}^{1-y} \sqrt{x^2 + y^2}\, dz\, dy\, dx\ .}$$

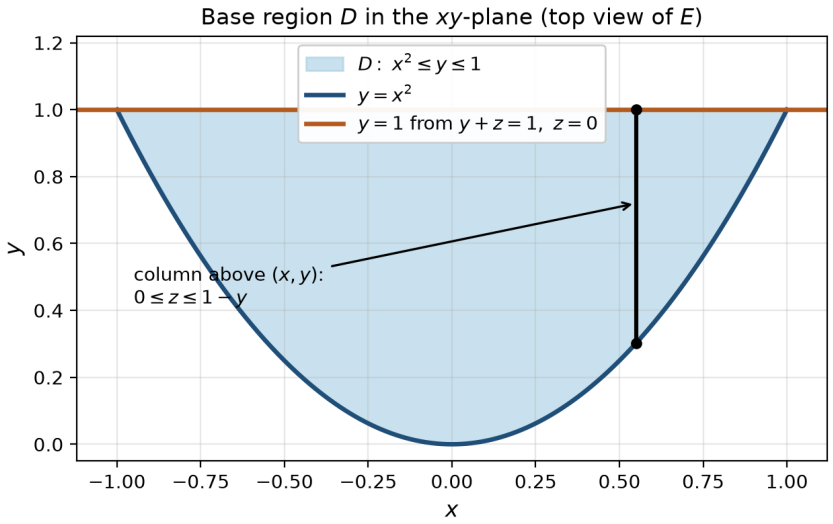


Figure 2: The base region D in the xy-plane, seen from above. The shaded set is $x^2 \le y \le 1$ with $-1 \le x \le 1$. The vertical segment is a typical column of E; it runs from $z = 0$ to $z = 1 - y$.

Part (b): the center of mass

The coordinates of the center of mass are the first moments divided by the mass:

$$\bar{x} = \frac{1}{m} \iiint_E x\rho \, dV, \qquad \bar{y} = \frac{1}{m} \iiint_E y\rho \, dV, \qquad \bar{z} = \frac{1}{m} \iiint_E z\rho \, dV.$$

The first moment with respect to the plane $x = 0$ is the integral with integrand $x\rho$. The other two moments use $y\rho$ and $z\rho$.

The $\bar{x}$ coordinate. The solid E is symmetric about the plane $x = 0$: the point $(-x, y, z)$ belongs to E exactly when (x, y, z) belongs to E. The density is even in x, because $\rho(-x, y, z) = \sqrt{x^2 + y^2} = \rho(x, y, z)$. The integrand $x\rho(x, y, z)$ is therefore odd in x. The integral of an odd function over the symmetric interval $-1 \le x \le 1$ is zero, so

$$\iiint_E x\rho \, dV = 0, \qquad \boxed{\bar{x} = 0 \, .}$$

The $\bar{y}$ coordinate.

$$\bar{y} = \frac{1}{m} \int_{-1}^{1} \int_{x^2}^{1} \int_{0}^{1-y} y\sqrt{x^2 + y^2} \, dz \, dy \, dx.$$

Because the integrand does not contain z,

$$\int_{0}^{1-y} y\sqrt{x^2 + y^2} \, dz = (1 - y)\, y\sqrt{x^2 + y^2}.$$

Hence

$$\boxed{\bar{y} = \frac{1}{m} \int_{-1}^{1} \int_{x^2}^{1} y(1 - y)\sqrt{x^2 + y^2} \, dy \, dx}$$

with m from part (a).

The $\bar{z}$ coordinate. The inner integral now contains z:

$$\int_0^{1-y} z\sqrt{x^2+y^2}\,dz = \sqrt{x^2+y^2}\Big[\frac{z^2}{2}\Big]_0^{1-y} = \frac{(1-y)^2}{2}\sqrt{x^2+y^2}.$$

Therefore

$$\boxed{\bar{z} = \frac{1}{m}\int_{-1}^{1}\int_{x^2}^{1}\frac{(1-y)^2}{2}\sqrt{x^2+y^2}\,dy\,dx}$$

with m from part (a). The center of mass is

$$\boxed{(\bar{x},\bar{y},\bar{z}) = \Big(0,\ \frac{M_{xz}}{m},\ \frac{M_{xy}}{m}\Big)}$$

where

$$M_{xz} = \int_{-1}^{1}\int_{x^2}^{1} y(1-y)\sqrt{x^2+y^2}\,dy\,dx, \qquad M_{xy} = \int_{-1}^{1}\int_{x^2}^{1}\frac{(1-y)^2}{2}\sqrt{x^2+y^2}\,dy\,dx.$$

Part (c): the moment of inertia about the z-axis

The distance of the point (x,y,z) from the z-axis is $\sqrt{x^2+y^2}$. The moment of inertia about the z-axis is the integral of the squared distance times the density:

$$I_z = \iiint_E (x^2+y^2)\,\rho(x,y,z)\,dV.$$

On E the density equals $\sqrt{x^2+y^2}$, so the integrand is $(x^2+y^2)^{3/2}$:

$$I_z = \int_{-1}^{1}\int_{x^2}^{1}\int_0^{1-y}(x^2+y^2)^{3/2}\,dz\,dy\,dx.$$

The integrand does not contain z, so

$$\int_0^{1-y}(x^2+y^2)^{3/2}\,dz = (1-y)(x^2+y^2)^{3/2}.$$

Hence

$$\boxed{I_z = \int_{-1}^{1}\int_{x^2}^{1}(1-y)(x^2+y^2)^{3/2}\,dy\,dx}$$

or, with all three integrations displayed,

$$\boxed{I_z = \int_{-1}^{1}\int_{x^2}^{1}\int_0^{1-y}(x^2+y^2)^{3/2}\,dz\,dy\,dx\ .}$$

A numerical remark (not part of the requested answer)

The exercise asks only for the expressions. To confirm that the limits are correct, the same integrals were evaluated numerically. Table 1 lists the results. The volume $V = 8/15$ is exact and follows from $\int_{-1}^{1}\int_{x^2}^{1}(1-y)\,dy\,dx$. The other entries are numerical values, and they are not closed forms.

Two independent numerical methods give the same values to six digits: a Gauss–Legendre product rule and a Monte Carlo average on E. The step scripts `part1_step1.py`, `part1_step2.py`, `part2_step1.py` and `part3_step1.py` print `PASS` for every comparison.

Quantity	Expression	Value
Volume V	$\int_{-1}^{1}\int_{x^2}^{1}(1-y)\,dy\,dx$	$8/15 = 0.533333$
Mass m	part (a)	0.295835
$\bar{x}$	part (b)	0
$\bar{y}$	part (b)	0.533509
$\bar{z}$	part (b)	0.233245
I_z	part (c)	0.158648

Table 1: Numerical checks of the set-up. The mass is less than the volume, because $\rho = \sqrt{x^2+y^2} \le \sqrt{2}$ on E and the density is smaller than 1 near the z-axis. The value $\bar{x} = 0$ follows from symmetry, not from a numerical integration.

How to solve problems of this kind

1. **Draw the solid before you write any integral.** Identify each bounding surface and the curve where two surfaces meet. In this exercise the curve $y = 1$, $z = 0$ is easy to miss, and it fixes the top of the base region.

2. **Find the projection first.** Project E on a coordinate plane in which the description is simple. Here the projection on the xy-plane is the region between a parabola and a line, so x and y have constant or one-variable limits.

3. **Choose the order that makes the inner limits simple.** A vertical line meets E in one segment, from $z = 0$ to $z = 1 - y$. Thus z is the inner variable. If the solid were not z-simple, you would split the integral into several pieces.

4. **Test the limits with a point.** Take a point that you know is inside E, for example $(0, 1/2, 1/4)$. It satisfies $x^2 \le y \le 1$ and $0 \le z \le 1 - y$. Take a point that you know is outside, for example $(0, 1/2, 3/4)$, and check that it violates $z \le 1 - y$.

5. **Use symmetry for the moments.** When the solid and the density are symmetric about a plane, the corresponding first moment is zero. This saves one full integration and gives an exact answer.

6. **Check the dimensions.** The mass has the dimension of density times volume. The center of mass has the dimension of length. The moment of inertia has the dimension of mass times length squared. A formula with the wrong dimension is wrong.

7. **When the problem says "do not evaluate", still check the limits.** Evaluate the volume, or evaluate the full integral numerically. A correct set-up must reproduce the volume. If the numerical value of the volume disagrees with the exact value, the limits are wrong.